LOGBOOK OF FMU_explore DEVELOPMENT

Here I describe the status of development the module FMU_explore. The code has been developed over long time as a gernal part of a setup script for BPL applications. Now the time has come to encapsulate this code as a central module. The simplified FMU interaction can be used for Modelica simulation models in general and not just BPL applications.

This informal text briefly describes the status and goals in near time. The idea is that this text here will complement the formal git-version handling. Now and then I plan to introduce reference to the git-version commit history.

Key architecture features are:

- Class structure of the module
- The module is then adapted to an applications through a number of variables
- The module placed in a package to simplify future use both locally and in Github

Development is mainly done in the following files:

- fmu_explore_pyfmi.py
- fmu_explore_fmpy.py
- setup TEST2_Batch.py
- setup TEST2_Batch_explore.py

I think the different examples serve as a good test bench for changes in the library. The collection of processes also is a good starting point when new systems are developed.

A general goal is:

Specific goals - now reached:

- Module v1 without simu(), show(), setLines() / Done 260819
- Module v2 complete a new concept axes as a list / Done 260819
- Module v2 now works in Windows with PyFMI / Done 260820
- Module v2 now works in Linux with PyFMI and no surprises / Done 260821
- Module v2 stored in package and used both locally and in Colab / Done 260824
- How do I make the performance function cstrProdMax() available inside simu() / Done

260825

- Make sim_res available in the Jupyter/Colab notebook when needed / Done 260829
- Make simulationTime default for simu() and also possible to set in simu() / Done 260901
- Make options in simu() possible to set / Done 260901
- Make cstrProdMax() available during plotting / Done 260901
- Make profile() available during plotting / Done 260901
- Fix simu('cont') for fmu_explore_fmpy and model_get() / Done 260907
- Make declaration of version in only one place from todays two places / Done 260909
- Made prevFinalTime local to the FMU_explore class / Done 260909
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- **Make version of scipy.optimize visible in system_info()**
- Reorganise the list of arguments for the class (and update notebooks)
- Handle cstrProdMax etc for fmpy

Day to day notes in reversed time order

Note that commit ID are now given after the date and description (i.e. opposite to Git). Far from all commits are described here though.

2026-09-08 Fixed in fmu_explore_fmpy/decribe_gemeral() a model.get to model_get. Call it now **ver 1.1.6**.

2026-09-08 Now BPL_TEST2_Fedbatch and BPL_YEAST_AIR_Fedbatch_DOcontrol works with fmpy in ver 1.1.5.

2026-09-07 Now BPL_TEST2_Batch works with fmu*explore*fmpy in **ver 1.1.5**. This version of FMU_explore will be uploaded to Github.

2026-09-02 Start the work with fmu_explore_fmpy and call it ver 1.1.5

2026-09-01 Brought in the liast three applictions Chemostat, Perfusion and IEC_validation and using **FMU_explore ver 1.1.4**.

2026-09-01 Brought in BPL_TEST2_Batch_calibration.

2026-09-01 I made simu() have parameters simultionTime and options free to set but default taken from the class. Call it **FMU_explorer ver 1.1.3**.

2026-08-28 Now brought in BPL_YEAST_COB_Batch and BPL_TEST2_Batch_design_space using FMU_explore ver 1.1.2

2026-08-29 I made sim_res available outside simu() by just introducting a return statement in the function. Call this **FMU_explorer ver 1.1.2**.

2026-08-28 Now brought in BPL_IEC_Operation, BPL_STEM_AIR_Perfusion.

2026-08-27 Now brought in BPL_CHO_Perfusion*cspr*openloop.

2026-08-26 Now brought three more applications to Colab: PL_TEST2_Chemostat, BPL_YEAST_AIR_Fedbatch, BPL_CHO_Fedbatch.

2026-08-25 Almost managed to convert BPL_TEST2_Chemostat for FMU_explore 1.1.0. Remains to handle cstrProdMax() defined in user functions, though.

2026-08-24 The package updated to include import of pandas. Then run locally with BPL_TEST2_Batch, BPL_TEST2_Fedbatch and BPL_CHO_Fedbatch in Windows. The next step is to run these applications also in Colab after updating the Colab notebook. Problem today with pyfmi more general.

2026-08-21 I tested the application BPL_TEST2_Batch in Linux and worked fine. Then I made it public at Github Colab and published a note as well.

2026-08-20 I decided to let the itertool work from the module only. The function resetPen() can be used in the function newplot() in the applications script and that is the key. Also the sim_res file is for the time being just kept global from the module.

2026-08-19 Managed to get v2 working reasonbly. The key was to introduce an empty list ax, similar to diagrams, as part of the class. Then newplot() gives both of them appropirate values. Remains to fix setLines() and also simu() bring back start and final simulation time.